Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: M. & M. Moules Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
12951 E Harney LN	Lodi	San Joaquin	95240

Street and nearest cross street (if no address): Jack Tone Rd (cross street)

Date facility was originally placed in operation: 03/01/1993

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X063-X250-XX10-XXXX

B. OPERATORS

Moules, Melvin

Operator name: Moules, Melvin

Telephone no.: (209) 333-1976 Landline, (209) 747-7376 Cellular

Mailing Address Number and Street	City	State	Zip Code
14117 N Jack Tone RD	Lodi	CA	95240

This operator is responsible for paying permit fees.

C. OWNERS

Machado, Dina

Legal owner name: Machado, Dina

Telephone no.: (209) 333-1976 Landline, (209) 747-7376 Cellular

Mailing Address Number and Street	City	State	Zip Code
14124 N Jack Tone RD	Lodi	CA	95240

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	0	0	5	3	0	0
Maximum number	0	0	5	3	0	0
Average number	0	0	5	3	0	0
Avg live weight (lbs)	0	0	1,000	750		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 81.16 *tons per reporting period*

Total nitrogen from manure: 759.20 *lbs per reporting period*

After ammonia losses (30% loss applied): 531.44 *lbs per reporting period*

Total phosphorus from manure: 128.48 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 938,400 *gallons*

Total nitrogen generated: 131.56 *lbs*

Total phosphorus generated: 27.10 *lbs*

Total potassium generated: 306.97 *lbs*

Total salt generated: 1,409.57 *lbs*

	938,400	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	938,400	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
30 HP Well	Ground water
40 HP Well	Ground water
Mom's House	Ground water
New House	Ground water
Old House	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field A	7	7	1	both	X063-X250-XX10-XXXX
Field B	22	22	1	both	X063-X250-XX10-XXXX
Field C	10	10	1	both	X063-X250-XX11-XXXX
Field D	10	10	1	both	X063-X250-XX11-XXXX
Field E	22	22	1	both	X063-X250-XX10-XXXX X063-X250-XX11-XXXX
Field F	29	29	0	none	X063-X250-XX15-XXXX X063-X250-XX36-XXXX
Field G	15	15	0	none	X063-X250-XX36-XXXX
Field H	5	5	0	none	X063-X250-XX10-XXXX
Totals for areas that were used for application	71	71	5		
Totals for areas that were not used for application	49	49	0		
Land application area totals	120	120	5		

B. CROPS AND HARVESTS

Field A

Field name: Field A

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 7 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	189.00 *ton*	Dry-weight		69.0	10,200.00	2,810.00	9,440.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	45.90	178.20	0.00
Total actual harvest content	27.00	170.75	47.04	158.03	1,205.28

Field B

Field name: Field B

Field B

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 22 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	594.00 *ton*	Dry-weight		69.0	10,200.00	2,810.00	9,440.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	45.90	178.20	0.00
Total actual harvest content	27.00	170.75	47.04	158.03	1,205.28

Field C

Field name: Field C

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 10 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	270.00 *ton*	Dry-weight		69.0	10,200.00	2,810.00	9,440.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	45.90	178.20	0.00
Total actual harvest content	27.00	170.75	47.04	158.03	1,205.28

Field D

Field name: Field D

Field D

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 10 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	270.00 *ton*	Dry-weight		70.6	12,300.00	2,380.00	11,700.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	197.10	37.80	186.30	0.00
Total actual harvest content	27.00	195.27	37.78	185.75	1,047.82

Field E

Field name: Field E

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 22 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	594.00 *ton*	Dry-weight		70.6	12,300.00	2,380.00	11,700.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	197.10	37.80	186.30	0.00
Total actual harvest content	27.00	195.27	37.78	185.75	1,047.82

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field A - 06/01/2023: Corn, silage

Field name: Field A

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Scraped material	178.59	49.37	162.69	0.00	35.00 *ton*
Application event totals		178.59	49.37	162.69	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	5.78	0.00	0.00	0.00	864,000.00 *gal*
30 HP Well	Ground water	0.57	0.00	0.00	0.00	600,000.00 *gal*
Application event totals		6.34	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	2.04	0.42	4.77	21.89	102,000.00 *gal*
40 HP Well	Ground water	2.89	0.00	0.00	0.00	432,000.00 *gal*
30 HP Well	Ground water	0.28	0.00	0.00	0.00	300,000.00 *gal*
Application event totals		5.21	0.42	4.77	21.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	3.47	0.00	0.00	0.00	518,400.00 *gal*
30 HP Well	Ground water	0.34	0.00	0.00	0.00	360,000.00 *gal*
Application event totals		3.81	0.00	0.00	0.00	

Field A - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	3.47	0.00	0.00	0.00	518,400.00 *gal*
30 HP Well	Ground water	0.34	0.00	0.00	0.00	360,000.00 *gal*
Application event totals		3.81	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	3.47	0.00	0.00	0.00	518,400.00 *gal*
30 HP Well	Ground water	0.34	0.00	0.00	0.00	360,000.00 *gal*
Application event totals		3.81	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	3.47	0.00	0.00	0.00	518,400.00 *gal*
30 HP Well	Ground water	0.34	0.00	0.00	0.00	360,000.00 *gal*
Application event totals		3.81	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	3.47	0.00	0.00	0.00	518,400.00 *gal*
30 HP Well	Ground water	0.34	0.00	0.00	0.00	360,000.00 *gal*
Application event totals		3.81	0.00	0.00	0.00	

Field B - 06/01/2023: Corn, silage

Field name: Field B

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field B - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Scraped material	178.59	49.37	162.69	0.00	110.00 *ton*
Application event totals		178.59	49.37	162.69	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	3.77	0.00	0.00	0.00	1,771,200.00 *gal*
30 HP Well	Ground water	0.37	0.00	0.00	0.00	1,230,000.00 *gal*
Application event totals		4.14	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	3.31	0.00	0.00	0.00	1,555,200.00 *gal*
30 HP Well	Ground water	0.32	0.00	0.00	0.00	1,080,000.00 *gal*
Application event totals		3.63	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	1.82	0.37	4.25	19.50	285,600.00 *gal*
40 HP Well	Ground water	2.57	0.00	0.00	0.00	1,209,600.00 *gal*
30 HP Well	Ground water	0.25	0.00	0.00	0.00	840,000.00 *gal*
Application event totals		4.65	0.37	4.25	19.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	2.94	0.00	0.00	0.00	1,382,400.00 *gal*
30 HP Well	Ground water	0.29	0.00	0.00	0.00	960,000.00 *gal*
Application event totals		3.23	0.00	0.00	0.00	

Field B - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
30 HP Well	Ground water	0.29	0.00	0.00	0.00	960,000.00 *gal*
40 HP Well	Ground water	2.94	0.00	0.00	0.00	1,382,400.00 *gal*
Application event totals		3.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	2.94	0.00	0.00	0.00	1,382,400.00 *gal*
30 HP Well	Ground water	0.29	0.00	0.00	0.00	960,000.00 *gal*
Application event totals		3.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	2.94	0.00	0.00	0.00	1,382,400.00 *gal*
30 HP Well	Ground water	0.29	0.00	0.00	0.00	960,000.00 *gal*
Application event totals		3.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	2.94	0.00	0.00	0.00	1,382,400.00 *gal*
30 HP Well	Ground water	0.29	0.00	0.00	0.00	960,000.00 *gal*
Application event totals		3.23	0.00	0.00	0.00	

Field C - 06/01/2023: Corn, silage

Field name: Field C

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field C - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Scraped material	178.59	49.37	162.69	0.00	50.00 *ton*
Application event totals		178.59	49.37	162.69	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	5.06	0.00	0.00	0.00	1,080,000.00 *gal*
30 HP Well	Ground water	0.49	0.00	0.00	0.00	750,000.00 *gal*
Application event totals		5.55	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	1.86	0.38	4.34	19.92	132,600.00 *gal*
40 HP Well	Ground water	2.63	0.00	0.00	0.00	561,600.00 *gal*
30 HP Well	Ground water	0.26	0.00	0.00	0.00	390,000.00 *gal*
Application event totals		4.75	0.38	4.34	19.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	3.24	0.00	0.00	0.00	691,200.00 *gal*
30 HP Well	Ground water	0.32	0.00	0.00	0.00	480,000.00 *gal*
Application event totals		3.55	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	3.24	0.00	0.00	0.00	691,200.00 *gal*
30 HP Well	Ground water	0.32	0.00	0.00	0.00	480,000.00 *gal*
Application event totals		3.55	0.00	0.00	0.00	

Field C - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	3.24	0.00	0.00	0.00	691,200.00 *gal*
30 HP Well	Ground water	0.32	0.00	0.00	0.00	480,000.00 *gal*
Application event totals		3.55	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	3.24	0.00	0.00	0.00	691,200.00 *gal*
30 HP Well	Ground water	0.32	0.00	0.00	0.00	480,000.00 *gal*
Application event totals		3.55	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	3.24	0.00	0.00	0.00	691,200.00 *gal*
30 HP Well	Ground water	0.32	0.00	0.00	0.00	480,000.00 *gal*
Application event totals		3.55	0.00	0.00	0.00	

Field D - 06/01/2023: Corn, silage

Field name: Field D

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Scraped material	178.59	49.37	162.69	0.00	50.00 *ton*
Application event totals		178.59	49.37	162.69	0.00	

Field D - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	5.06	0.00	0.00	0.00	1,080,000.00 *gal*
30 HP Well	Ground water	0.49	0.00	0.00	0.00	750,000.00 *gal*
Application event totals		5.55	0.00	0.00	0.00	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	1.86	0.38	4.34	19.92	132,600.00 *gal*
40 HP Well	Ground water	2.63	0.00	0.00	0.00	561,600.00 *gal*
30 HP Well	Ground water	0.26	0.00	0.00	0.00	390,000.00 *gal*
Application event totals		4.75	0.38	4.34	19.92	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	3.24	0.00	0.00	0.00	691,200.00 *gal*
30 HP Well	Ground water	0.32	0.00	0.00	0.00	480,000.00 *gal*
Application event totals		3.55	0.00	0.00	0.00	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	3.24	0.00	0.00	0.00	691,200.00 *gal*
30 HP Well	Ground water	0.32	0.00	0.00	0.00	480,000.00 *gal*
Application event totals		3.55	0.00	0.00	0.00	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	3.24	0.00	0.00	0.00	691,200.00 *gal*
30 HP Well	Ground water	0.32	0.00	0.00	0.00	480,000.00 *gal*
Application event totals		3.55	0.00	0.00	0.00	

Field D - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	3.24	0.00	0.00	0.00	691,200.00 *gal*
30 HP Well	Ground water	0.32	0.00	0.00	0.00	480,000.00 *gal*
Application event totals		3.55	0.00	0.00	0.00	

09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	3.24	0.00	0.00	0.00	691,200.00 *gal*
30 HP Well	Ground water	0.32	0.00	0.00	0.00	480,000.00 *gal*
Application event totals		3.55	0.00	0.00	0.00	

Field E - 06/05/2023: Corn, silage

Field name: Field E

Crop: Corn, silage

Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Scraped material	178.59	49.37	162.69	0.00	110.00 *ton*
Application event totals		178.59	49.37	162.69	0.00	

06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	3.77	0.00	0.00	0.00	1,771,200.00 *gal*
30 HP Well	Ground water	0.37	0.00	0.00	0.00	1,230,000.00 *gal*
Application event totals		4.14	0.00	0.00	0.00	

Field E - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	1.82	0.37	4.25	19.50	285,600.00 *gal*
40 HP Well	Ground water	2.57	0.00	0.00	0.00	1,209,600.00 *gal*
30 HP Well	Ground water	0.25	0.00	0.00	0.00	840,000.00 *gal*
Application event totals		4.65	0.37	4.25	19.50	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	2.94	0.00	0.00	0.00	1,382,400.00 *gal*
30 HP Well	Ground water	0.29	0.00	0.00	0.00	960,000.00 *gal*
Application event totals		3.23	0.00	0.00	0.00	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	2.94	0.00	0.00	0.00	1,382,400.00 *gal*
30 HP Well	Ground water	0.29	0.00	0.00	0.00	960,000.00 *gal*
Application event totals		3.23	0.00	0.00	0.00	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	2.94	0.00	0.00	0.00	1,382,400.00 *gal*
30 HP Well	Ground water	0.29	0.00	0.00	0.00	960,000.00 *gal*
Application event totals		3.23	0.00	0.00	0.00	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	2.94	0.00	0.00	0.00	1,382,400.00 *gal*
30 HP Well	Ground water	0.29	0.00	0.00	0.00	960,000.00 *gal*
Application event totals		3.23	0.00	0.00	0.00	

Field E - 06/05/2023: Corn, silage

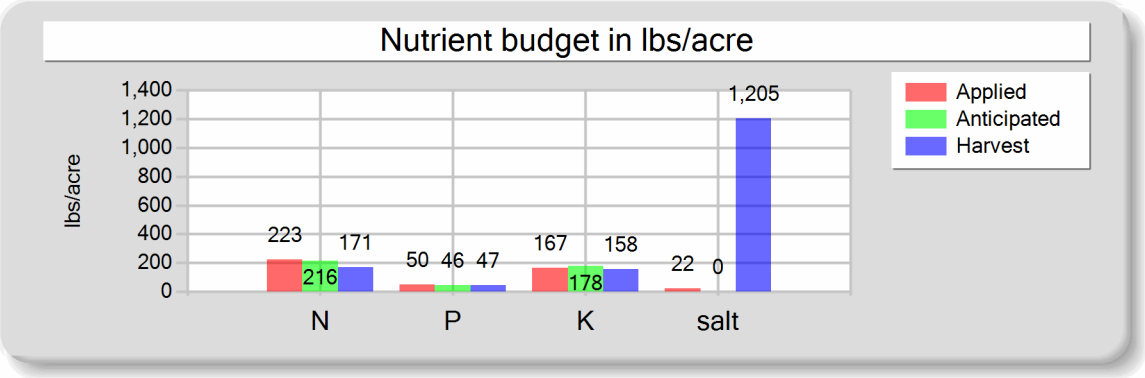
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
40 HP Well	Ground water	2.94	0.00	0.00	0.00	1,382,400.00 *gal*
30 HP Well	Ground water	0.29	0.00	0.00	0.00	960,000.00 *gal*
Application event totals		3.23	0.00	0.00	0.00	

B. NUTRIENT BUDGET

Field A - 06/01/2023: Corn, silage

Field name: Field A Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	178.59	49.37	162.69	0.00
Process wastewater	2.04	0.42	4.77	21.89
Fresh water	28.55	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	223.17	49.79	167.46	21.89
Anticipated crop nutrient removal	216.00	45.90	178.20	0.00
Actual crop nutrient removal	170.75	47.04	158.03	1,205.28
Nutrient balance	52.43	2.75	9.43	-1,183.39
Applied to removed ratio	1.31	1.06	1.06	0.02

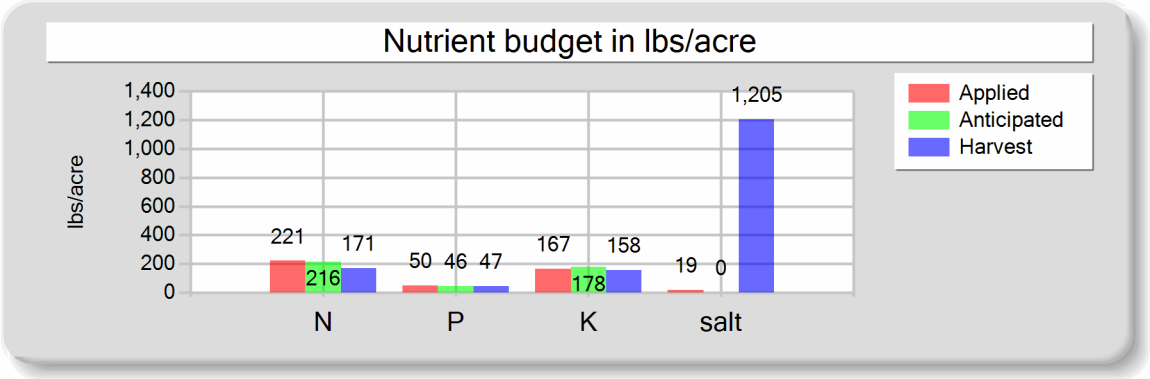
Fresh water applied	
6,588,000.00	*gallons*
242.61	*acre-inches*
34.66	*inches/acre*

Process wastewater applied	
102,000.00	*gallons*
3.76	*acre-inches*
0.54	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field B - 06/01/2023: Corn, silage

Field name: Field B Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	178.59	49.37	162.69	0.00
Process wastewater	1.82	0.37	4.25	19.50
Fresh water	26.74	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	221.15	49.74	166.94	19.50
Anticipated crop nutrient removal	216.00	45.90	178.20	0.00
Actual crop nutrient removal	170.75	47.04	158.03	1,205.28
Nutrient balance	50.40	2.70	8.91	-1,185.78
Applied to removed ratio	1.30	1.06	1.06	0.02

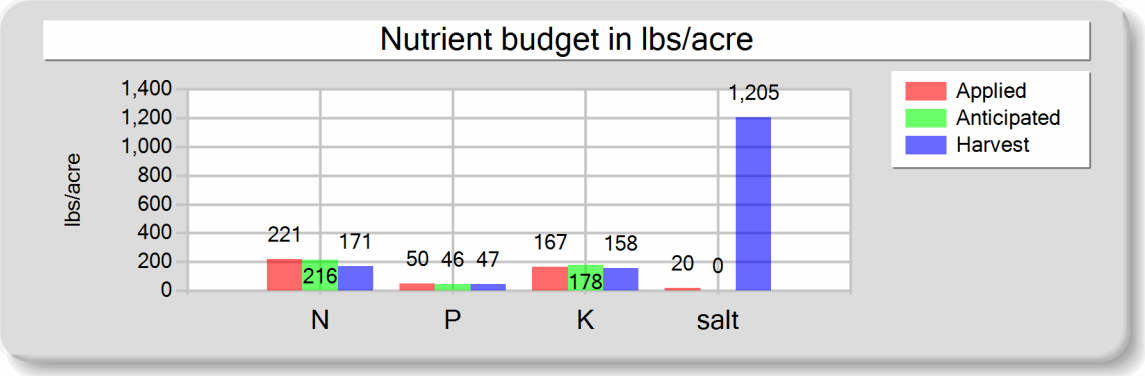
Fresh water applied	
19,398,000.00	*gallons*
714.36	*acre-inches*
32.47	*inches/acre*

Process wastewater applied	
285,600.00	*gallons*
10.52	*acre-inches*
0.48	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field C - 06/01/2023: Corn, silage

Field name: Field C Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	178.59	49.37	162.69	0.00
Process wastewater	1.86	0.38	4.34	19.92
Fresh water	26.20	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	220.64	49.75	167.03	19.92
Anticipated crop nutrient removal	216.00	45.90	178.20	0.00
Actual crop nutrient removal	170.75	47.04	158.03	1,205.28
Nutrient balance	49.89	2.71	9.00	-1,185.36
Applied to removed ratio	1.29	1.06	1.06	0.02

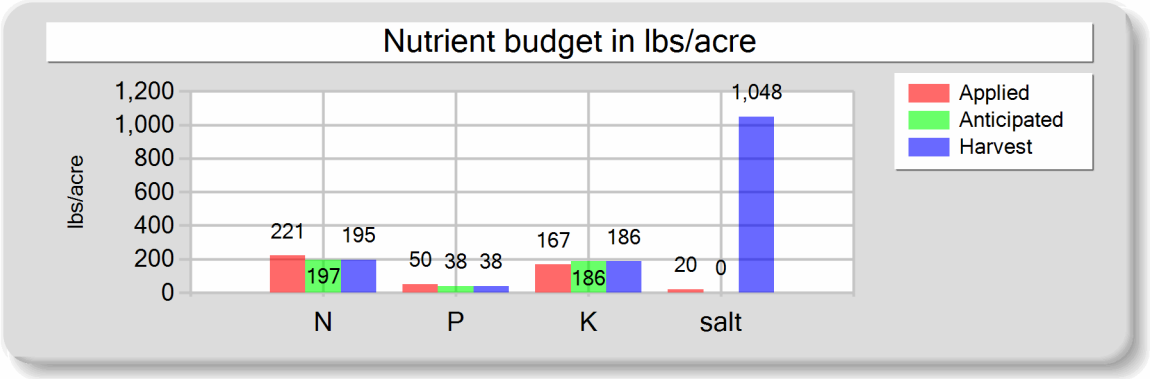
Fresh water applied	
8,637,600.00	*gallons*
318.09	*acre-inches*
31.81	*inches/acre*

Process wastewater applied	
132,600.00	*gallons*
4.88	*acre-inches*
0.49	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field D - 06/01/2023: Corn, silage

Field name: Field D Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	178.59	49.37	162.69	0.00
Process wastewater	1.86	0.38	4.34	19.92
Fresh water	26.20	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	220.64	49.75	167.03	19.92
Anticipated crop nutrient removal	197.10	37.80	186.30	0.00
Actual crop nutrient removal	195.27	37.78	185.75	1,047.82
Nutrient balance	25.37	11.97	-18.72	-1,027.90
Applied to removed ratio	1.13	1.32	0.90	0.02

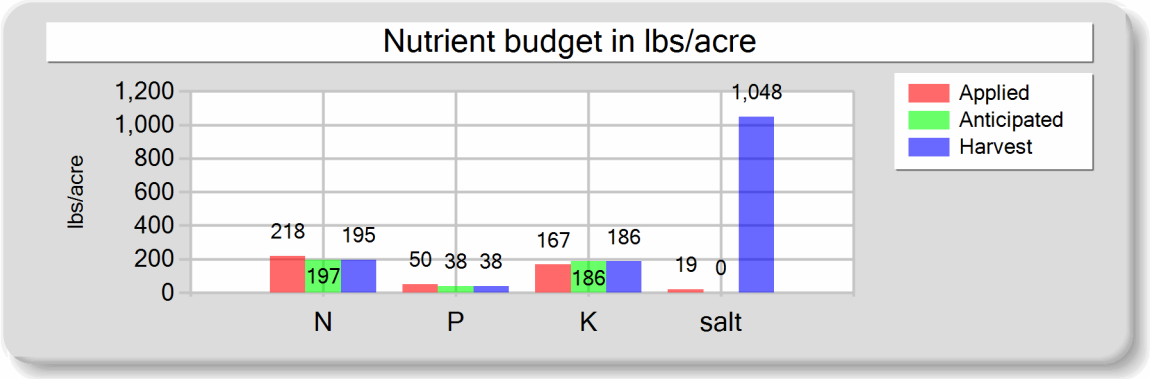
Fresh water applied	
8,637,600.00	*gallons*
318.09	*acre-inches*
31.81	*inches/acre*

Process wastewater applied	
132,600.00	*gallons*
4.88	*acre-inches*
0.49	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field E - 06/05/2023: Corn, silage

Field name: Field E Crop: Corn, silage Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	178.59	49.37	162.69	0.00
Process wastewater	1.82	0.37	4.25	19.50
Fresh water	23.11	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	217.52	49.74	166.94	19.50
Anticipated crop nutrient removal	197.10	37.80	186.30	0.00
Actual crop nutrient removal	195.27	37.78	185.75	1,047.82
Nutrient balance	22.24	11.96	-18.81	-1,028.32
Applied to removed ratio	1.11	1.32	0.90	0.02

Fresh water applied	
16,762,800.00	*gallons*
617.32	*acre-inches*
28.06	*inches/acre*

Process wastewater applied	
285,600.00	*gallons*
10.52	*acre-inches*
0.48	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

T3132037B

Sample and source description: T3132037B

Sample date: 05/12/2023 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,100.00	5,280.00	17,400.00							0.00
DL	1.00	1.00	1.00							1.00

T3264051A

Sample and source description: T3264051A

Sample date: 09/21/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 9.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,400.00	5,080.00	12,300.00	20,200.00	7,160.00	1,440.00	6,270.00	1,240.00		50.90
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

B. PROCESS WASTEWATER ANALYSES

T3053008A

Sample and source description: T3053008A

Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	67.20	8.40	0.00	0.00	10.90	146.00								793.00	360
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3132037A

Sample and source description: T3132037A

Sample date: 05/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	16.80	5.60	0.00	0.00	3.46	39.20								434.00	180
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

C. FRESH WATER ANALYSES

30 HP Well

T2320012B

Sample description: T2320012B

Sample date: 11/16/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.79	0.00	0.79								216.00	0
DL	0.50	0.50	0.50								100.00	1

40 HP Well

T2320012A

Sample description: T2320012A

Sample date: 11/16/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.61	0.00	5.61								576.00	0
DL	1.00	0.50	1.00								100.00	1

Mom's House

Mom's House

T3314018A

Sample description: T3314018A

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.55	0.00	5.55								427.00	0
DL	1.00	0.50	1.00								100.00	1

New House

T3314018C

Sample description: T3314018C

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.74	0.00	1.74								228.00	0
DL	1.00	0.50	1.00								100.00	1

Old House

T3314018B

Sample description: T3314018B

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.43	0.00	3.43								412.00	0
DL	1.00	0.50	1.00								100.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field A - 06/01/2023: Corn, silage

T3291031A

Sample and source description: T3291031A

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,200.00	2,810.00	9,440.00		7.20
DL	1.00	1.00	1.00		1.00

Field B - 06/01/2023: Corn, silage

T3291031A

Sample and source description: T3291031A

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,200.00	2,810.00	9,440.00		7.20
DL	1.00	1.00	1.00		1.00

Field C - 06/01/2023: Corn, silage

Field C - 06/01/2023: Corn, silage

T3291031A

Sample and source description: T3291031A

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,200.00	2,810.00	9,440.00		7.20
DL	1.00	1.00	1.00		1.00

Field D - 06/01/2023: Corn, silage

T3291031B

Sample and source description: T3291031B

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,300.00	2,380.00	11,700.00		6.60
DL	1.00	1.00	1.00		1.00

Field E - 06/05/2023: Corn, silage

T3291031B

Sample and source description: T3291031B

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,300.00	2,380.00	11,700.00		6.60
DL	1.00	1.00	1.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

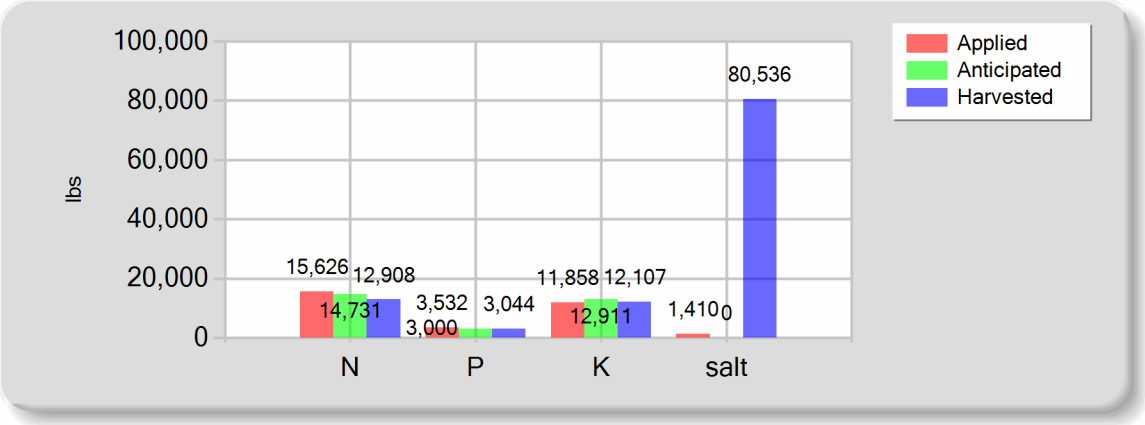
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

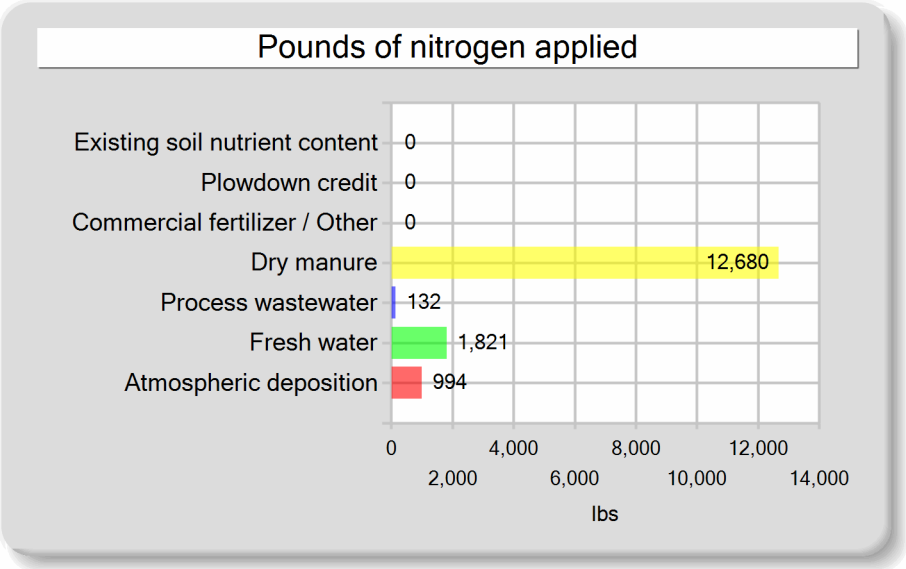
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

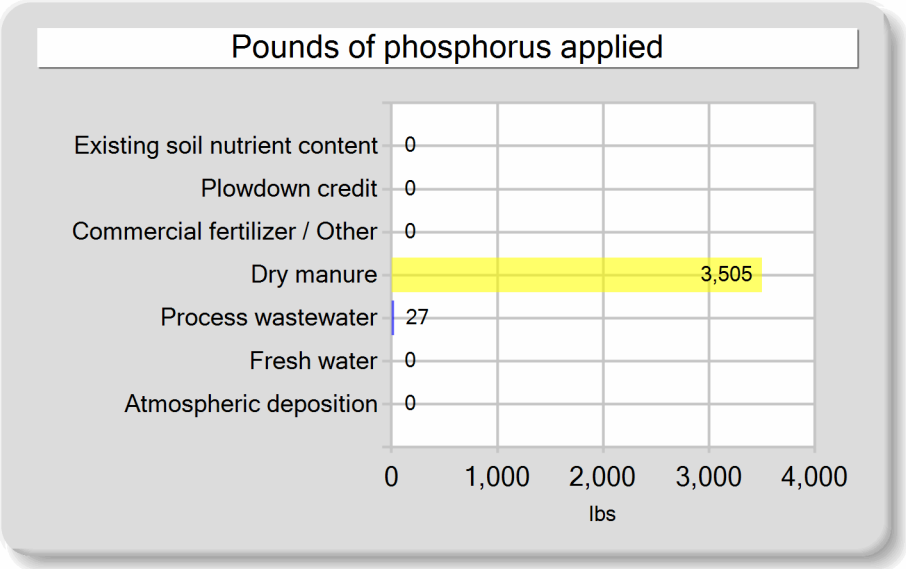
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	12,679.54	3,505.13	11,550.99	0.00
Process wastewater	131.56	27.10	306.97	1,409.57
Fresh water	1,820.57	0.00	0.00	0.00
Atmospheric deposition	994.00	0.00	0.00	0.00
Total nutrients applied	15,625.66	3,532.22	11,857.96	1,409.57
Anticipated crop nutrient removal	14,731.20	2,999.70	12,911.40	0.00
Actual crop nutrient removal	12,907.97	3,043.65	12,106.97	80,536.03
Nutrient balance	2,717.70	488.57	-249.01	-79,126.46
Applied to removed ratio	1.21	1.16	0.98	0.02

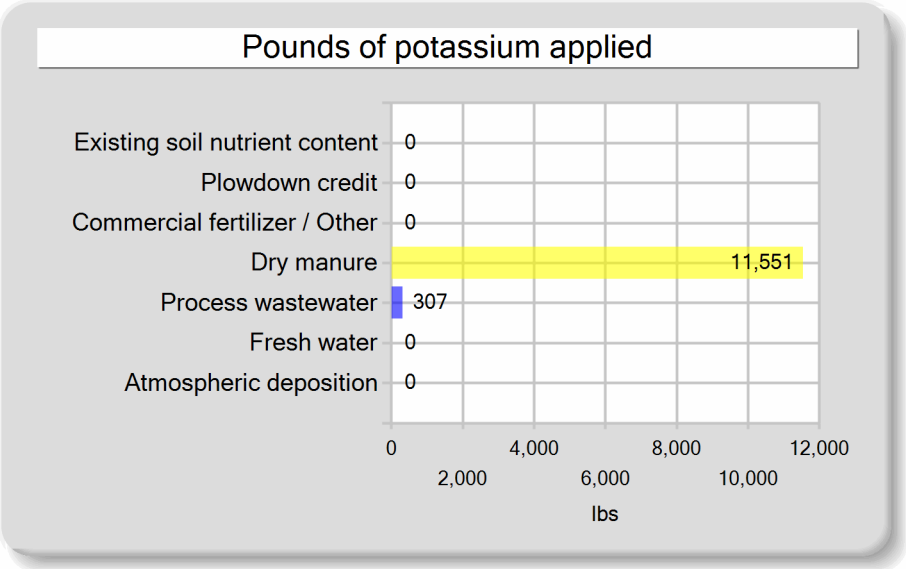
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

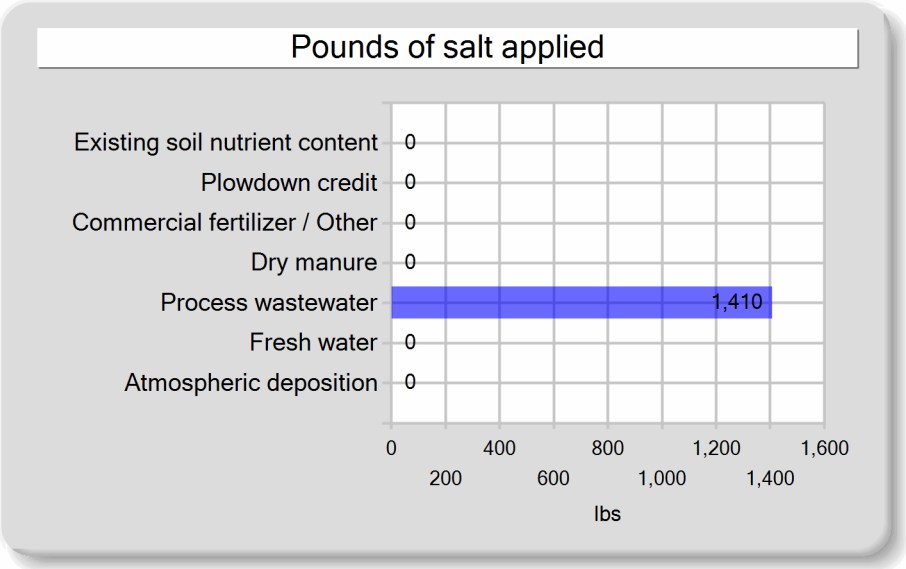


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Dina Machado
SIGNATURE OF OWNER OF FACILITY

Dina Machado
PRINT OR TYPE NAME

6-18-24
DATE

Melvin Moules
SIGNATURE OF OPERATOR OF FACILITY

Melvin Moules
PRINT OR TYPE NAME

6-18-24
DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Authorized Copy Released To:
JCRK, INC.
(209) 327-0992
Sampled By: Madalena Moules
Project ID:

Client: M & M MOULES DAIRY
Mailing address: 14117 N. JACK TONE RD.
(not facility address) LODI, CA 95240
Phone#: (209) 333-1976

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	11/9/23	8:00a	Mom's House	Dom. well	11.2c	20 min	X				
2	✓	8:30a	Old House	✓	11.0	✓	X				
3	✓	9:00a	New House	✓	10.8	✓	X				
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	Jacquie Captein	JCRK		11/10/23 8:00a		
2nd		DENELE ANALYTICAL, INC.	11-10-23 1:00			T3314018

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
M & M Moules Dairy
14117 N. Jack Tone Rd.
Lodi, CA 95240

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/14/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Mom's House	T3314018A	Electrical Conductivity (EC)	EPA 120.1 and	0.427	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	5.55	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	273	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.6			
Electrical Conductivity (EC)	93.2			1.39
Nitrate Nitrogen (NO3-N)	102	100 (T3314015D)	100.0 (T3314015D)	0.459

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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Compliance Analysis Report

Customer
M & M Moules Dairy
14117 N. Jack Tone Rd.
Lodi, CA 95240

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/14/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Old House	T3314018B	Electrical Conductivity (EC)	EPA 120.1 and	0.412	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	3.43	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	264	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.6			
Electrical Conductivity (EC)	93.2			1.39
Nitrate Nitrogen (NO3-N)	102	100 (T3314015D)	100.0 (T3314015D)	0.459

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Compliance Analysis Report

Customer
M & M Moules Dairy
14117 N. Jack Tone Rd.
Lodi, CA 95240

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/14/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 10.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
New House	T3314018C	Electrical Conductivity (EC)	EPA 120.1 and	0.228	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.74	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	146	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.6			
Electrical Conductivity (EC)	93.2			1.39
Nitrate Nitrogen (NO3-N)	102	100 (T3314015D)	100.0 (T3314015D)	0.459

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ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

1232 SOUTH AVENUE
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Authorized Copy Released To: JCRK, INC.
(209) 327-0992

Client: M & M MOULES DAIRY
Mailing address: 14117 N. JACK TONE RD.
(not facility address) LODI, CA 95240
Phone#: (209) 333-1976

Sampled By: Madelena Moules
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH3-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH3-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	11/15/22	9:00a	40 HP Well	Ag Well	15.6°	20 min.	X				
2	✓	9:30a	30 HP Well	✓	14.8°	✓	X				
3	✓	10:00a	Old House Domestic	Domestic Well	16.2°	✓					H14
4	✓	10:30a	New House Dom.	✓	15.8°	✓	X				
5	✓	11:00a	Mom's House Dom.	✓	15.4°	✓	X				
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
1st	Jacquie Capteon	JCRK		11/16/22 8:00a		
2nd		DENELE ANALYTICAL, INC.	11-16-22 12:40			T2320012

(LABORATORY USE ONLY)

Denele Analytical, Inc.

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Compliance Analysis Report

Customer
M & M Moules Dairy
14117 N. Jack Tone Rd.
Lodi, CA 95240

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/16/22 12:40 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/17/22

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 15.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
40 HP Well	T2320012A	Electrical Conductivity (EC)	EPA 120.1 and	0.576	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	5.61	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	368	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	93.4	93.4 (T2320012A)	98.8 (T2320012A)	5.61
Electrical Conductivity (EC)	102			0.830
Nitrate Nitrogen (NO3-N)	96.2	94.3 (T2315012A)	97.8 (T2315012A)	3.60

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ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
M & M Moules Dairy
14117 N. Jack Tone Rd.
Lodi, CA 95240

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/16/22 12:40 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/17/22

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 14.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
30 HP Well	T2320012B	Electrical Conductivity (EC)	EPA 120.1 and	0.216	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.788	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	138	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	93.4	93.4 (T2320012A)	98.8 (T2320012A)	5.61
Electrical Conductivity (EC)	102			0.830
Nitrate Nitrogen (NO3-N)	96.2	94.3 (T2315012A)	97.8 (T2315012A)	3.60

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.